



Workshop: Late Deliveries

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi
dell'Aquila

Two hours, one dataset, six steps

- 1,200 parcels from a courier company: which ones arrive late?
- You take them through the whole workflow of lessons 1 to 5
- Each step: fill the blanks ____, run the cell, run the check
- The check prints ✓, or ✗ with a hint
- Each step ends with one question, answered in a sentence

The six steps

Step	What you do
1	look at the data, and compute the baseline
2	split, keeping the share of late parcels equal
3	one pipeline: fill the gaps, scale, logistic regression
4	cross-validation, on the training set only
5	could a curve do better? Same folds
6	test the winner, once

Before you start

- Start the course container and open `Workshop/workshop.ipynb`
- Run the first cell: it loads the data
- `NameError: name '___' is not defined` means a blank is still empty
- Do **not** open `delivery_data.py`: it holds the answer

How did it go?

- Hands up for each step that printed ✓
- Where the crosses were, and which lesson that step came from
- The $\pm$ of step 4 and step 5: is the difference real?

The rule, revealed

- Longer routes, fuller vans and rain make a parcel **later**
- Driver experience makes it **earlier**
- Departure time works through rush hour: the further from 13:00, the later
- The age of the van has **no effect at all**